

Design Audit Agent

Aivar Associate ML Engineer Technical Assessment Submission

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Introduction

User Interface (UI) and User Experience (UX) design play a crucial role in determining how effectively users interact with digital products. Modern applications require interfaces that are visually appealing, accessible, consistent, and intuitive. Traditionally, evaluating interface quality involves manual reviews performed by designers and developers, which can be time-consuming and subjective.

With recent advancements in multimodal artificial intelligence, vision-language models can analyze visual content and reason about interface design in a manner similar to human reviewers. This creates an opportunity to automate portions of the design review process while still providing meaningful and actionable feedback.

The Design Audit Agent was developed to address this challenge. It is an AI-powered system that leverages Google Gemini 2.5 Flash Vision to analyze screenshots and generate structured design evaluations. The application supports both single-image design auditing and before-and-after regression analysis, helping developers and designers quickly identify design issues and interface changes.

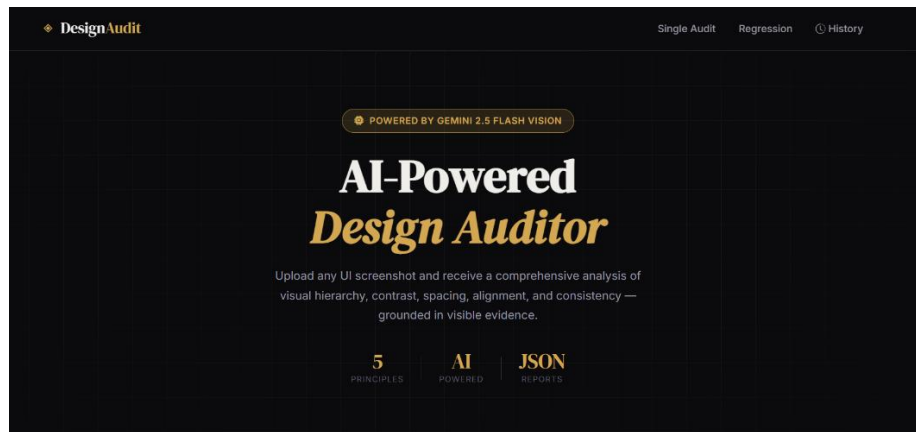


Figure 1: AI-Powered Design Audit Agent Interface

Problem Statement

Design reviews are an essential part of software development, yet they often require significant manual effort. As applications evolve through multiple iterations, maintaining visual consistency and identifying unintended design regressions become increasingly difficult.

Small modifications introduced during development may negatively affect readability, spacing, accessibility, alignment, or overall user experience. Existing automated tools typically focus on pixel differences rather than providing meaningful design feedback.

The objective of this project is to develop an intelligent system capable of analyzing screenshots, identifying design issues, generating actionable recommendations, and producing structured reports that support design decision-making.

Solution Overview

The Design Audit Agent consists of two primary modules designed to address different stages of the design review process.

The first module, Single Page Design Analysis (Level 1), accepts a screenshot and evaluates it against five core design principles: visual hierarchy, contrast, spacing, alignment, and consistency. The system identifies potential issues and provides recommendations for improvement along with confidence scores and severity classifications.

The second module, Before-and-After Regression Analysis (Level 2), accepts a baseline screenshot and a modified version of the same interface. The system compares both screenshots and identifies visual changes, classifying them as improvements, regressions, or neutral modifications. An overall verdict is then generated to summarize the impact of the changes.

By combining these two capabilities, the application provides both proactive design evaluation and change-based quality assurance.

Technology Stack

The application was developed using Python and Flask for backend development. Google Gemini 2.5 Flash Vision was used as the multimodal AI engine responsible for screenshot analysis and reasoning. The frontend interface was built using HTML, CSS, and JavaScript, while generated reports are stored using lightweight JSON-based persistence.

Core Technologies:

Component	Technology
Backend	Python 3.10, Flask
AI Model	Gemini 2.5 Flash Vision
Frontend	HTML, CSS, JavaScript
Storage	JSON-based Report Storage

System Architecture

The solution follows a modular architecture that separates user interaction, AI analysis, report processing, and data storage responsibilities.

Google Gemini 2.5 Flash Vision serves as the core reasoning engine, enabling the system to understand UI layouts, identify design issues, and generate structured design recommendations.

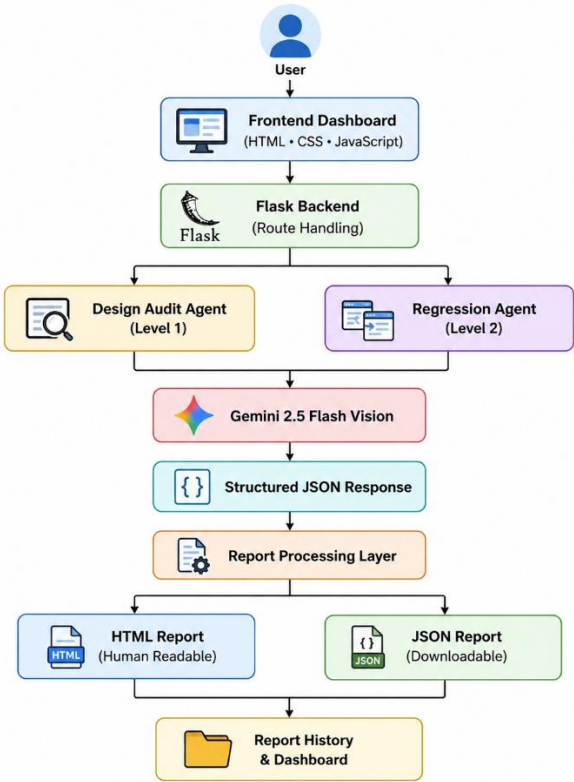


Figure 2: High-Level Architecture of the Design Audit Agent

The frontend dashboard handles file uploads and displays generated reports. The Flask backend manages request processing, validation, and communication with Gemini Vision. Responses generated by the AI model are converted into structured JSON before being processed by the reporting layer. Reports are stored locally and can be accessed later through the report history functionality.

Features and Implementation

The application was designed to satisfy all requirements of Level 1 and Level 2 while maintaining a clean and user-friendly experience.

The Single Page Design Analysis module evaluates uploaded screenshots against five design principles. For each detected issue, the system generates detailed findings that include the violated principle, severity level, confidence score, issue location, user impact, and a specific recommendation. Severity levels are categorized as Critical, High, Medium, Low, and Informational, enabling users to prioritize improvements effectively.

The Before-and-After Regression Analysis module compares two versions of the same interface and identifies visual differences between them. Each detected change is classified as an improvement, regression, or neutral modification. The generated report explains what changed, where the change occurred, its impact on user experience, and the confidence associated with the observation.

Additional features implemented in the application include downloadable JSON reports, report history management, confidence indicators, severity badges, and a responsive dashboard interface.

The application has been deployed on Render, allowing users to access and evaluate the system through a publicly available web interface without requiring local installation or configuration.

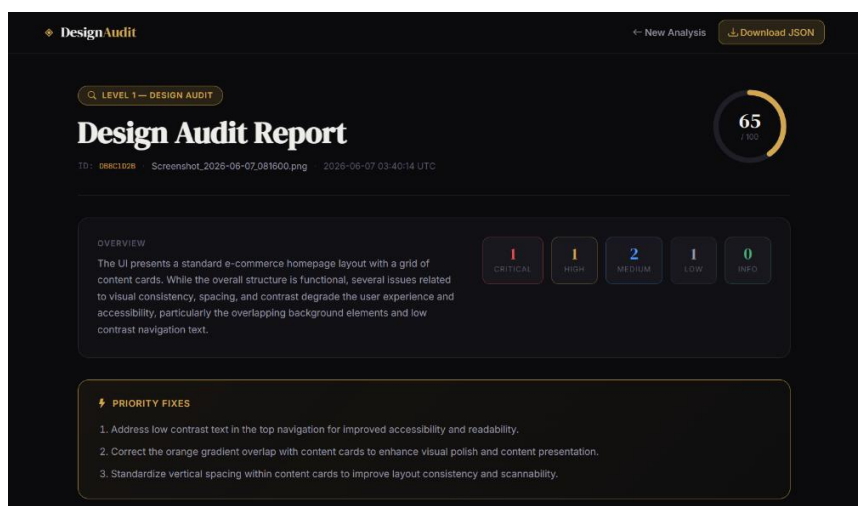


Figure 3: Sample Design Audit Report

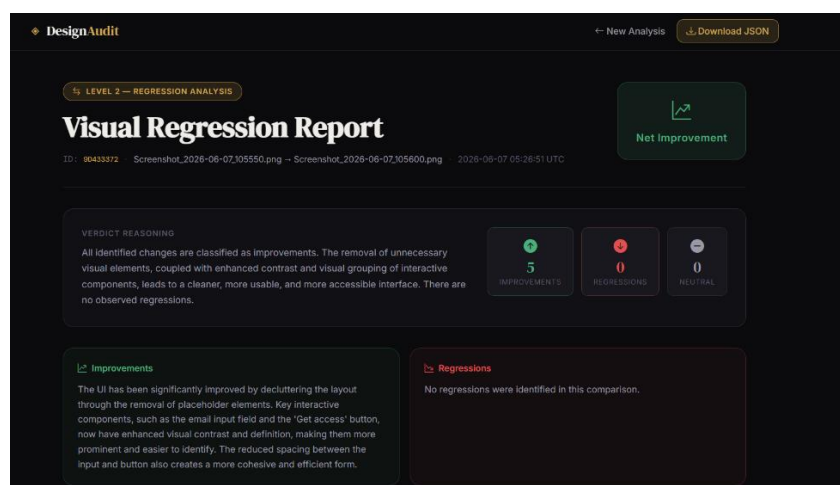


Figure 4: Sample Regression Analysis Report

Challenges and Design Decisions

One of the primary challenges encountered during development was ensuring that the AI model consistently returned structured responses. Since large language models can occasionally produce unstructured output, prompt engineering and response validation techniques were used to enforce a predictable JSON schema.

Another challenge involved reducing hallucinated observations. Vision-language models may sometimes infer information that is not directly visible in an image. To address this issue, the prompts explicitly instructed the model to generate only evidence-based findings that could be verified within the uploaded screenshots.

Special attention was also given to report presentation. Raw JSON outputs are difficult for users to interpret, so a dedicated reporting interface was developed to convert technical outputs into visually understandable reports. Confidence scores and severity indicators were introduced to improve transparency and help users prioritize issues.

The application uses low-temperature model settings to improve consistency and reduce variability in generated results. Additionally, lightweight JSON storage was selected instead of a traditional database in order to simplify deployment while maintaining report persistence.

Results

The developed system successfully implements all required functionalities for both Level 1 (Single Page Design Analysis) and Level 2 (Before-and-After Regression Analysis) of the assessment.

The Design Audit module generates structured findings containing design principles, issue locations, user impacts, recommendations, severity classifications, and confidence scores. The Regression Analysis module successfully identifies visual differences between interface versions and classifies them as improvements, regressions, or neutral changes.

The application also provides report persistence, downloadable JSON outputs, and a modern dashboard interface for interacting with generated analyses. Testing was performed using multiple UI screenshots and before-and-after interface variations to verify the correctness of issue detection, change classification, report generation, and dashboard functionality.

The generated reports enable users to quickly identify critical design issues, understand the impact of interface changes, and prioritize improvements through structured and actionable feedback.

Conclusion and Future Scope

The Design Audit Agent successfully demonstrates the application of multimodal AI for UI/UX evaluation and visual regression analysis. By combining Google Gemini 2.5 Flash Vision with structured reporting and an intuitive dashboard, the system provides actionable design insights that can assist developers and designers in improving interface quality. The project satisfies Level 1 and Level 2 challenge requirements while maintaining a practical and user-friendly implementation.

Future enhancements may include automatic accessibility scoring based on WCAG guidelines, pixel-level visual comparison for more precise regression detection, PDF report exports, browser extension support, and CI/CD integration for automated visual testing during software deployment pipelines.

The complete source code, implementation details, and project assets are available through the links provided below.

Project Links

Live Demo: <https://design-audit-agent.onrender.com/>

GitHub Repository: <https://github.com/Dinesh-NPC/Design-Audit-Agent>

LinkedIn Profile: <https://www.linkedin.com/in/dinesh-kumar-kct/>